Exhibits (tables and figures) for: Revealing hidden jump strategies beyond conventional force-plate metrics

Tables

Table 1

Sample characteristics

Variable	*M*	*SD*	*Mdn*	Min	Max
Panel A. Athletes (n = 212)					
Age at first test (years)	16.0	5.9	14.1	8.8	37.9
Body mass at first test (kg)	53.3	15.9	51.7	25.0	96.0
Trials per athlete	3.8	2.3	3.0	1.0	16.0
Sessions per athlete	1.5	0.8	1.0	1.0	6.0
Panel B. Trials (n = 805)					
Jump height (cm)	28.2	8.4	27.0	8.5	51.5
Concentric peak force (BW)	2.35	0.38	2.29	1.65	4.70
Peak landing force (BW)	5.39	1.97	4.97	2.19	14.88
Eccentric braking impulse (N s)	27.8	15.4	24.6	0.0	95.2
Concentric impulse, first 100 ms (N s)	62.2	26.7	58.7	1.5	187.9
Take-off duration (ms)[a]	782	190	774	306	1484
Landing duration (ms)[b]	247	114	235	70	928

Note. Values describe the analysed sample after quality control. Decimal places are fixed within each variable. *M* = mean; *Mdn* = median; BW = multiples of body weight. Trials were recorded at 1000 Hz (746) or 500 Hz (59). [a]Movement onset to take-off. [b]Touch-down to the lowest position of the centre of mass.

Table 2

Regions of each phase where force was associated with jump height, age and body mass

Predictor	Region (% of phase)[a]	Peak (%)	Effect	95% CI
Panel A. Take-off, effect in BW per SD of the predictor				
Jump height	14–30, 66–68, 72–84, 88–98	95	0.25	[0.22, 0.29]
Age	98	98	0.09	[0.06, 0.12]
Body mass	2–6, 98–100	98	−0.06	[−0.09, −0.03]
Panel B. Landing, effect in BW per SD of the predictor				
Jump height	4–28	14	0.36	[0.21, 0.52]
Age	–	–	–	–
Body mass	76–82	78	−0.18	[−0.30, −0.06]
Panel C. Jump height split within and between athletes, effect in BW per cm[b]				
Take-off, between athletes	18–29, 67–99	95	0.031	[0.026, 0.035]
Take-off, within athlete	3–29, 46–63, 92–99	58	0.029	[0.017, 0.042]
Landing, between athletes	9–15, 18–25	24	0.039	[0.016, 0.061]
Landing, within athlete	2–16, 23	13	0.115	[0.074, 0.164]

Note. Function-on-scalar regression of body-weight-normalised force on jump height, age and body mass, each standardised (1 *SD* = 8.4 cm, 5.7 years and 15.8 kg). Effect is the coefficient at the peak of the region, with a pointwise 95% confidence interval (CI) from 2000 bootstrap resamples of athletes. BW = multiples of body weight; dash = no region.

[a]Panels A and B: nodes where the simultaneous 95% bands of both the athlete-bootstrap least-squares fit and the functional mixed model (athlete and session random curves) exclude zero. [b]Athlete-mean jump height and the trial's deviation from it entered together; region from the athlete-bootstrap simultaneous band.

Table 3

Group differences in athlete-mean force curves

				Difference (BW)		Hedges' *g*	
Phase	Region (%)	*p*	Peak (%)	Value	95% CI	Value	95% CI
Panel A. Top vs bottom tertile of jump height (71 vs 71 athletes)							
Take-off	81–99	<.001	93	0.55	[0.48, 0.61]	2.63	[2.31, 2.94]
Landing	2–56	.003	15	0.75	[0.45, 1.04]	0.83	[0.50, 1.15]
Adjusted for age and body mass[a]							
Take-off	12–32	.021	–	–	–	–	–
	66–99	<.001	–	–	–	–	–
Landing	19–40	.043	–	–	–	–	–
Panel B. Aged 18 or over vs under 18 (48 vs 164 athletes)							
Take-off	87–99	<.001	96	0.47	[0.36, 0.56]	1.60	[1.25, 1.92]
Landing	none	.069[c]	–	–	–	–	–
Adjusted for body mass and jump height[a]							
Take-off	none	.123[c]	–	–	–	–	–
Landing	none	.226[c]	–	–	–	–	–

Note. One mean curve per athlete. Region = cluster of adjacent nodes (% of the phase) where the groups differed; *p* from 5000 permutations of the largest cluster mass, controlling the family-wise error rate over the phase[b]. Peak = node of the largest $|t|$ in the region, where the difference (first group minus second; BW = multiples of body weight) and Hedges' *g* are given with 95% confidence intervals (CIs) from 2000 bootstrap resamples of athletes within groups. Bonferroni threshold for the four primary tests: $p < .0125$.

[a]Freedman–Lane permutation of residuals. [b]Nodes with $|t|$ above the two-sided 5% threshold form clusters. [c]No significant region; smallest cluster *p* shown.

Table 4

Take-off and landing curve profiles: characteristics, evidence for distinct groups and stability

	Take-off profile		Landing profile	
	1	2	1	2
Panel A. Characteristics				
Trials (*n*)	329	476	560	245
Share of all trials (%)	41	59	70	30
Share of matched-height trials (%)[a]	35	65	74	26
Jump height (cm), mean[b]	29.6	27.2	28.4	27.7
Take-off duration (ms), mean	649	875	790	764
Landing duration (ms), mean	240	252	293	141
Panel B. Evidence for distinct groups				
Mean silhouette width (0 to 1)	0.30		0.37	
Groups preferred by the gap statistic[c]	1		6	
Dip test of unimodality, *p*	.741		.978	
Lowest bootstrap Jaccard similarity	0.93		0.97	
Panel C. Stability within athletes (83 athletes tested in two or more sessions)				
Same profile in every session (%)	55		70	
Expected by chance (%)	43		51	
Permutation *p*	.007		<.001	
Same profile for two trials in a session (%)	76		80	
Expected by chance (%)	52		58	
Agreement after elastic alignment[d]	–		0.17	

Note. Profiles from *k*-means clustering ($k = 2$) of functional principal component scores covering at least 90% of variance, separately for each phase. Dash = not applicable.

[a] Jump height between the 45th and 55th percentiles (26.0–28.2 cm; 82 trials). [b] Difference between profiles in a linear mixed model with an athlete random intercept: take-off $p < .001$, landing $p = .017$. [c] Candidates 1 to 6; 1 indicates no cluster structure. [d] Adjusted Rand index between profiles from aligned and unaligned landing curves (1 = identical, 0 = chance).

Figures

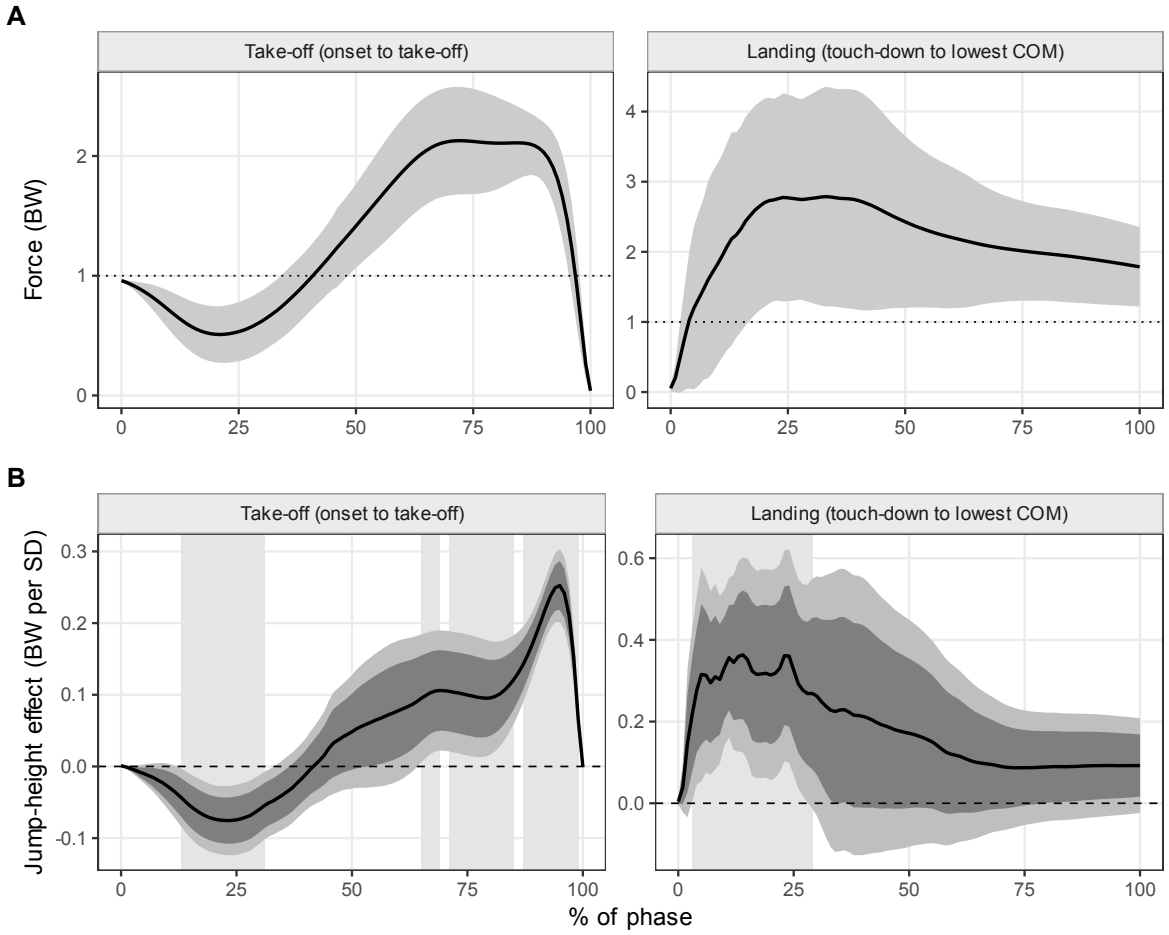


Fig. 1. Average curves and where jump height matters. (A) Mean force in multiples of body weight (BW) over the 805 analysed trials (line) with ± 1 SD (shaded); the dotted line is body weight. Take-off runs from movement onset to take-off and landing from touch-down to the lowest centre-of-mass (COM) position, each rescaled to 0–100%. (B) Jump-height coefficient from function-on-scalar regression (BW per SD of jump height, adjusted for age and body mass) with pointwise (dark) and simultaneous (light) 95% bands from resampling athletes. Pale background shading marks the regions where both estimation approaches agree.

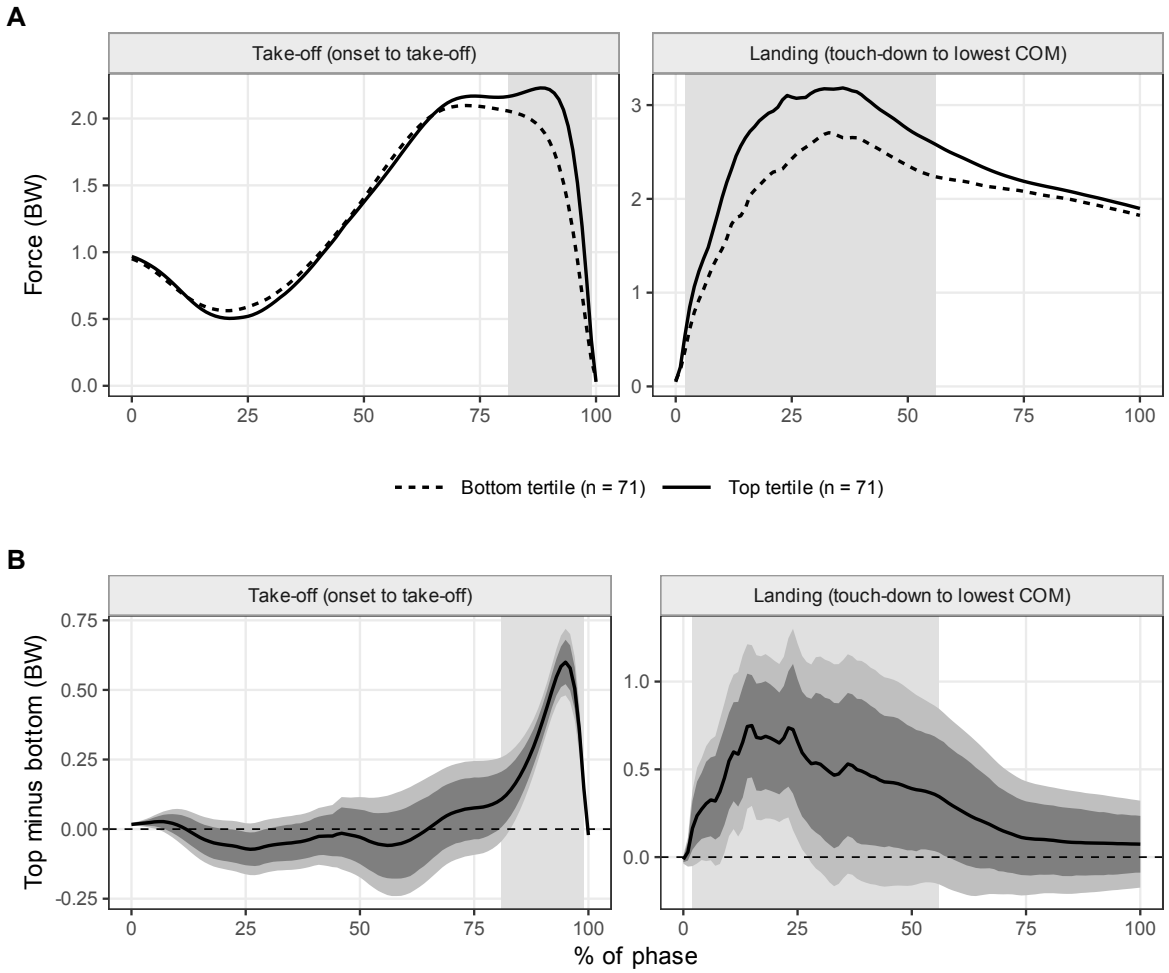


Fig. 2. Top versus bottom jump-height tertile, one mean curve per athlete. (A) Group mean curves. (B) Difference (top minus bottom) with pointwise (dark) and simultaneous (light) 95% bootstrap bands. Background shading marks clusters significant in the permutation test with family-wise error control.

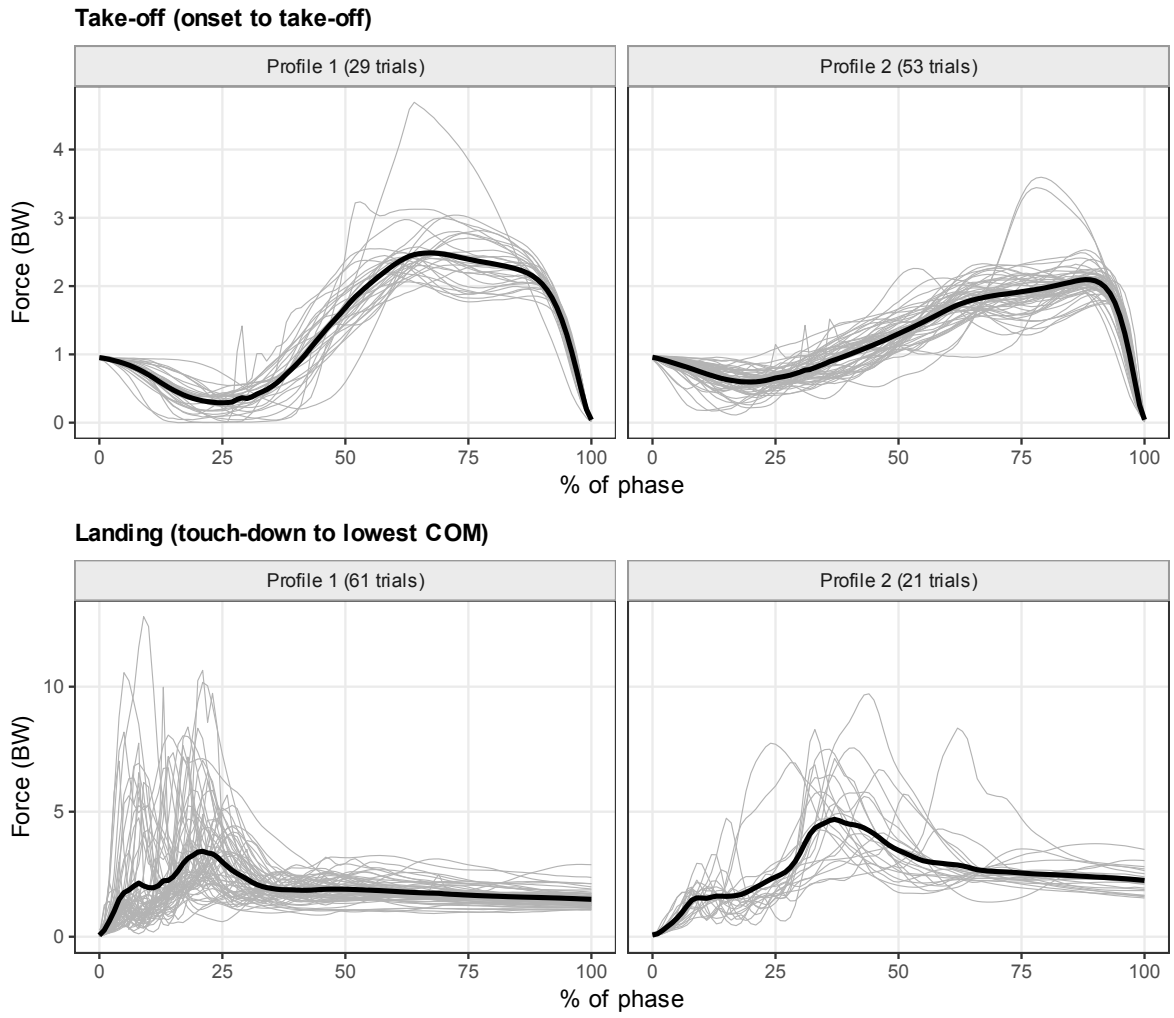


Fig. 3. Jumps of almost equal height (26.0–28.2 cm, 82 trials) still differ in shape. Thin grey lines are individual trials, split by curve profile; the thick black line is the profile mean. Top: take-off. Bottom: landing.

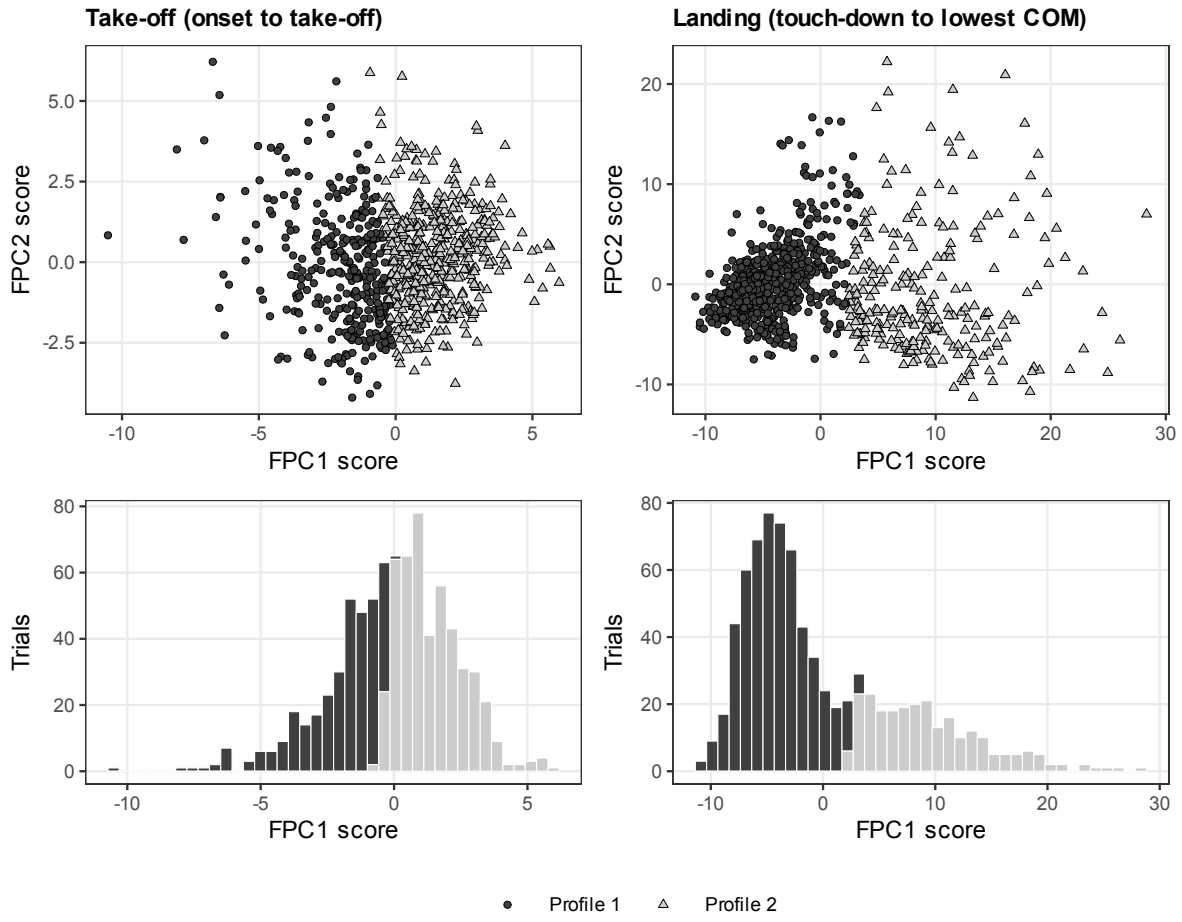


Fig. 4. Curve profiles form a continuum. For each phase, the scatter plot shows the first two functional principal component (FPC) scores of every trial, with symbols for the two k-means profiles, and the histogram shows the distribution of the first score shaded by profile. There is no gap between the profiles and no second mode.